```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/Placement .csv')
df.head()
```

	Student_ID	CGPA	IQ	Placement
0	1	6.8	123	1
1	2	5.9	106	0
2	3	5.3	121	0
3	4	7.4	132	1
4	5	5.8	142	0

Next steps: Generate code with df | New interactive sheet

```
plt.figure(figsize=(16,5))
plt.subplot(1,2,1)
sns.distplot(df['CGPA'])
plt.subplot(1,2,2)
sns.distplot(df['IQ'])
plt.show()
```

```
/tmp/ipykernel_7023/4237127157.py:3: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

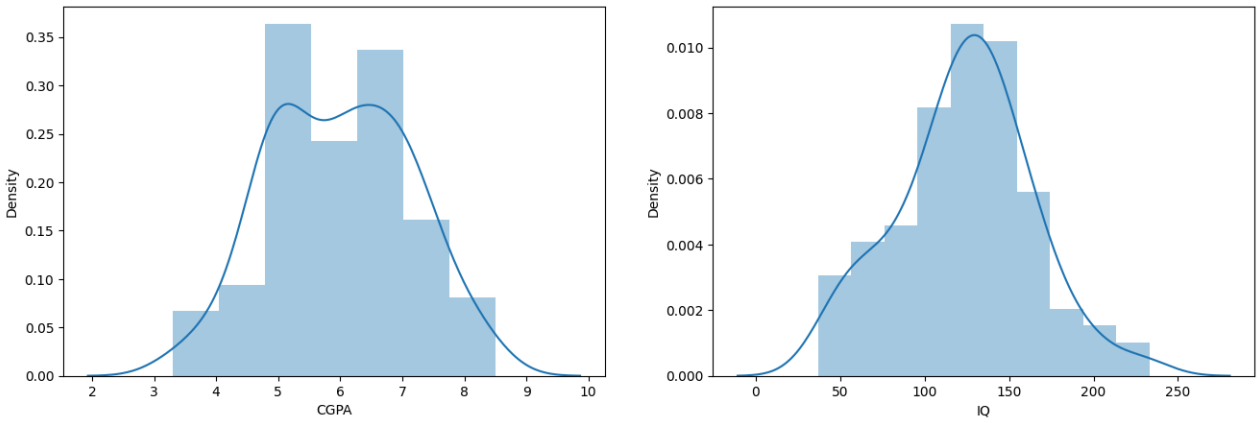
  sns.distplot(df['CGPA'])
/tmp/ipykernel_7023/4237127157.py:5: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

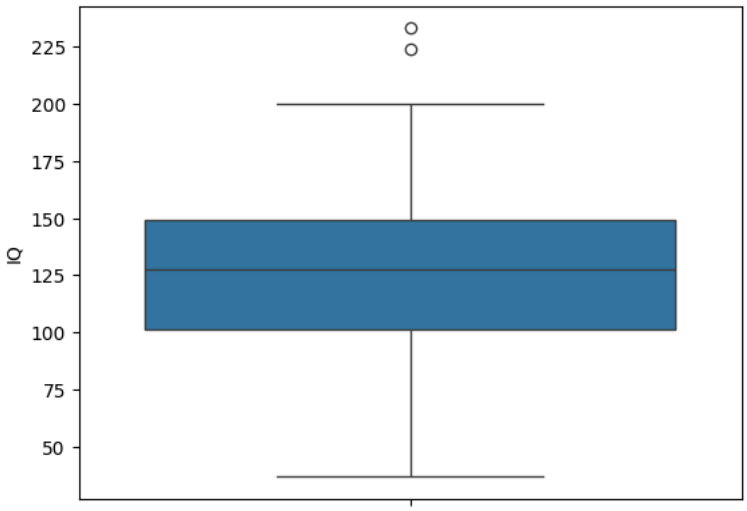
For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['IQ'])
```




```
sns.boxplot(df['IQ'])
```

```
<Axes: ylabel='IQ'>
```




```
percentile25 = df['IQ'].quantile(0.25)
percentile75 = df['IQ'].quantile(0.75)
```

```
# Calculate the interquartile range (IQR)
iqr = percentile75 - percentile25
```

```
upper_limit = percentile75 + 1.5 * iqr
lower_limit = percentile25 - 1.5 * iqr
```

```
df[df['IQ'] > upper_limit]
df[df['IQ'] < lower_limit]
```

Student_ID	CGPA	IQ	Placement

```
new_df = df[df['IQ'] < upper_limit]
new_df.shape
```

```
(98, 4)
```

..

```
plt.figure(figsize=(16,8))
plt.subplot(2,2,1)
sns.distplot(df['IQ'])
plt.subplot(2,2,2)
sns.boxplot(df['IQ'])
plt.subplot(2,2,3)
sns.distplot(new_df['IQ'])
plt.subplot(2,2,4)
sns.boxplot(new_df['IQ'])
plt.show()
```

```
/tmp/ipykernel_7023/3375542974.py:3: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

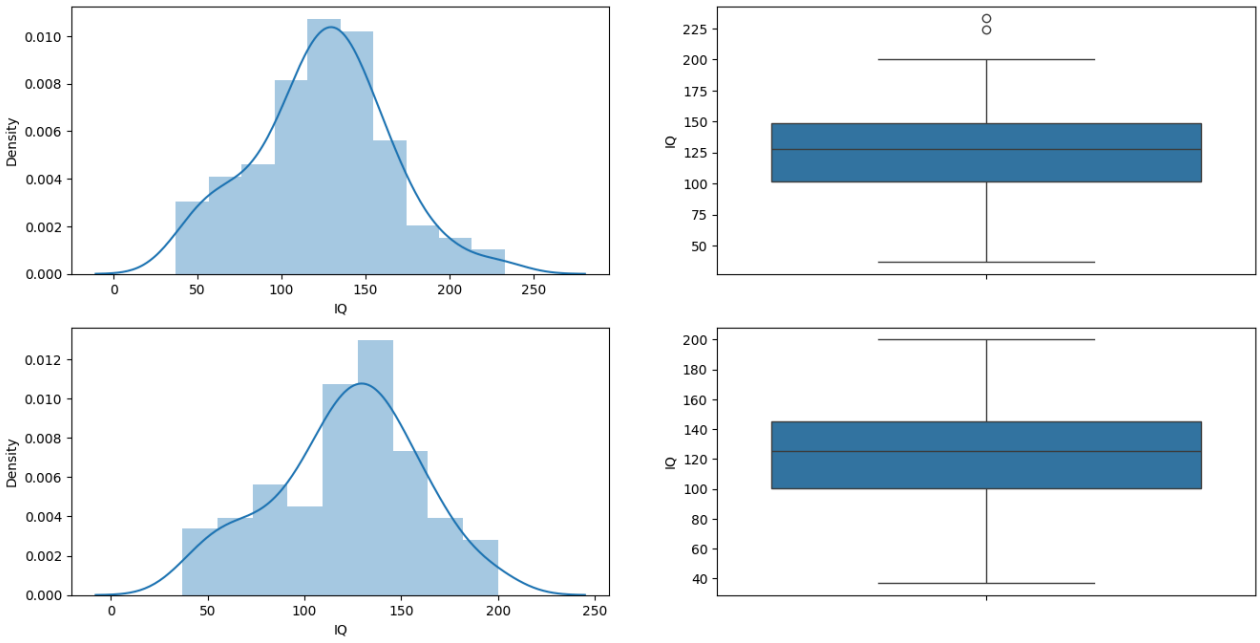
  sns.distplot(df['IQ'])
/tmp/ipykernel_7023/3375542974.py:7: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(new_df['IQ'])
```



Gemini

```
plt.figure(figsize=(16,8))
plt.subplot(2,2,1)
sns.distplot(df['IQ'])
sns.histplot(df['IQ'])
plt.subplot(2,2,2)
sns.boxplot(df['IQ'])
plt.subplot(2,2,3)
sns.distplot(new_df['IQ'])
sns.histplot(new_df['IQ'])
plt.subplot(2,2,4)
sns.boxplot(new_df['IQ'])
plt.show()
```

```
/tmp/ipykernel_7023/3375542974.py:3: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['IQ'])
/tmp/ipykernel_7023/3375542974.py:7: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(new_df['IQ'])
```

